

Task (a):

A MOD-10 counter has 10 states (0 to 9). With n flip-flops we get 2^n states, so we need $2^n \geq 10$. Since $2^3 = 8$ is too small and $2^4 = 16$ is enough, $n = 4$. So each MOD-10 counter uses 4 JK flip-flops.

A MOD-100 counter (00 to 99) is just two MOD-10 counters, one for the units digit and one for the tens digit. So the total is $4 + 4 = 8$ JK flip-flops.

In the asynchronous design, the tens stage gets its clock from the units stage instead of the main clock. When the units counter goes from 9 (1001) back to 0 (0000), that rollover triggers the tens counter, and the tens digit goes up by 1.

Task (b):

i) Each flip-flop has a delay of 10 ns. In a ripple counter the change passes through the flip-flops one by one, and in the worst case it goes through all 8 of them.

$$T_{\text{worst}} = 8 \times 10 \text{ ns} = 80 \text{ ns}$$

ii) The clock period cannot be shorter than the worst-case delay, so $T_{\text{min}} = 80 \text{ ns}$.

$$F_{\text{max}} = 1 / 80 \text{ ns} = 12.5 \text{ MHz}$$

iii) At 500 kHz the clock period is

$$T = 1 / (500 \times 10^3) \text{ s} = 2 \text{ } \mu\text{s} = 2000 \text{ ns}$$

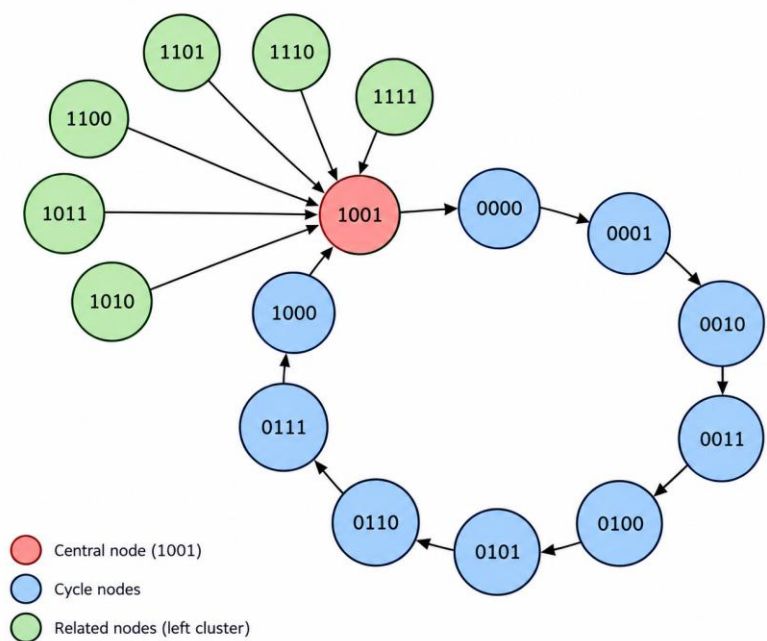
This is 25 times bigger than 80 ns, so the counter has enough time to settle before the next pulse. By timing alone, the count should be correct.

The flicker comes from the design itself. The flip-flops change one after another, so for a moment the output can show a wrong in-between value, and the display catches it. Noise on the sensor line in a crowded hall can also add false pulses.

Task (c):

Synchronous MOD-10 counter using JK flip-flops. All four flip-flops share one clock, and their J and K inputs come from the equations below.

State Diagram:



State sequence: 0 → 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 0

Excitation table for states 0 to 15:

State	Q3	Q2	Q1	Q0	Q3 ⁺	Q2 ⁺	Q1 ⁺	Q0 ⁺	J3	K3	J2	K2	J1	K1	J0	K0
0	0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
1	0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
2	0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
3	0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
4	0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
5	0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
6	0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
7	0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
8	1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
9	1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1
10	1	0	1	0	0	0	0	0	x	1	0	x	x	1	0	x
11	1	0	1	1	0	0	0	0	x	1	0	x	x	1	x	1
12	1	1	0	0	0	0	0	0	x	1	x	1	0	x	0	x
13	1	1	0	1	0	0	0	0	x	1	x	1	0	x	x	1
14	1	1	1	0	0	0	0	0	x	1	x	1	x	1	0	x
15	1	1	1	1	0	0	0	0	x	1	x	1	x	1	x	1

Equations from the table:

$$J_0 = 1, \quad K_0 = 1$$

$$J_1 = Q_0Q_3', \quad K_1 = Q_0 + Q_3$$

$$J_2 = Q_0Q_1Q_3', \quad K_2 = Q_0Q_1 + Q_3$$

$$J_3 = Q_0Q_1Q_2, \quad K_3 = Q_0 + Q_1 + Q_2$$

Task (d):

i) Both digits use the same clock. The tens digit is only allowed to count when the units digit is at 9 (1001). This is told by the terminal count signal:

$$TC = Q_3 \cdot Q_0$$

When $TC = 1$, the tens digit counts on the next clock edge. When $TC = 0$, it holds its value. Since all flip-flops change on the same clock edge, there is no ripple delay and no glitch between the two digits.

ii) The flip-flop delay counts only once now. I assumed the gate delay for TC is 10 ns and the setup time is 5 ns.

$$T = T_{FF} + T_{TC} + T_S = 10 + 10 + 5 = 25 \text{ ns}$$

$$F_{\max} = 1 / 25 \text{ ns} = 40 \text{ MHz}$$

This is about 3.2 times faster than the 12.5 MHz of the asynchronous design, and there are no glitches.

Task (e):

Asynchronous: the simplest and uses the least hardware, but the delays add up and glitches can appear when counting fast.

Synchronous: needs more gates, but all flip-flops change together, so it is fast and clean.

Hybrid: sits in between, but there is still a delay (and possible glitches) between the digits.

For the hall, I would choose the fully synchronous MOD-100 design. It can run up to 40 MHz while the input is only 500 kHz, and it removes the ripple that caused the flicker.